LIVER GENOMICS · SINGLE-NUCLEUS RE-ANALYSIS

No transcriptional de-zonation signal in matched MASLD biopsies

— but the pericentral detox program dims

A critical re-analysis of the single-nucleus RNA-seq in Gribben et al., Nature 2024 (GSE202379)

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47 donors · ~99,809 nuclei · 69,426 hepatocytes · raw-count, donor-level analysis

BACKGROUND

Hepatocytes are transcriptionally zonated

Pericentral (zone 3)	Periportal (zone 1)

Central vein

Portal tract

GLUL, CYP3A4, CYP2E1 — drug metabolism (Wnt-driven)

ASS1, CPS1, PCK1, HAL, ALDOB — urea cycle

THE PUBLISHED CLAIM — Gribben 2024

“Hepatocytes lose their zonation.”

- End-stage hepatocytes co-express pericentral + periportal markers — validated by imaging + protein.
- Hepatocyte↔cholangiocyte transdifferentiation, “prominent in end-stage.”

OUR QUESTION

Does hepatocyte zonation degrade across MASLD — and is it linked to the hepatocyte→cholangiocyte transdifferentiation they describe?

The catch

our first approach is misleading — and the tissue itself is confounded

A relative ruler — and what it hides

THE OLD RULER

We first read it with a z-scored zonation coordinate and marker–marker correlations, pooled across all samples.

WHAT IT SHOWED

"Hepatocytes lose their zonation; the pericentral program turns off, strongest at end-stage."

This is the published finding — our first, relative reading reproduced it.

THE TRAP

Hides who moved

no counts — which cells, how many, and from where.

Mixes the samples

pools healthy + biopsy + end-stage across a hidden acquisition break.

Bent by artifacts

depth, cell number and tissue source all tilt a z-score or correlation.

→ so we count molecules instead.

Stage is entangled with how the tissue was taken

① The two extremities are rouge - acquisition isn't needle biopsy

Healthy	F0	F1	F2	F3	F4	End-stage
deceased-donor	16-gauge needle biopsies					explants

The ends aren't acquisition-matched to the biopsies (2/8/12/12/4) → **we use biopsy-only F1–F4.**

② And they also carry organ-wide stress

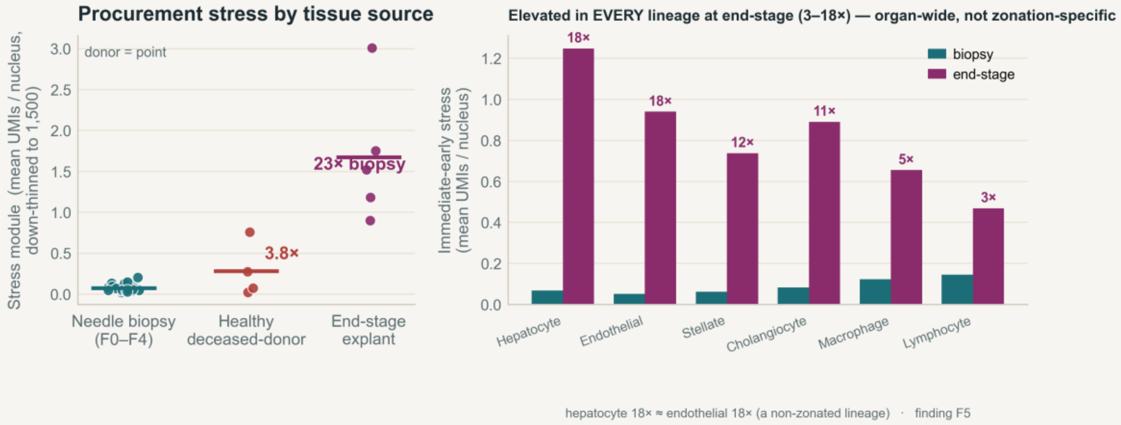


Figure 1. Procurement-stress module by tissue source (donor-level).

Stress spikes even where there's no zonation.

Immediate-early / heat-shock genes jump 18× in the ends — but **endothelium, which has no zonation program, jumps 18× too**; hypoxia (HIF) stays flat. So it's not necessary zonation change affect – but rather, at least as probable - acute handling[1].

[1] *Dissociation / handling artifact: van den Brink et al. 2017; O'Flanagan et al. 2019; Denisenko et al. 2020. Provenance: paper Methods + GSE202379 metadata.*

Method

how we re-test it, in counts

Four classes — and the four ways they can move

Every hepatocyte is one of four classes (left).
"Losing zonation" = cells redistribute across them — four distinct moves (right), each with its own count signature.

THE FOUR CLASSES

periportal program →

	PP –	PP +
PC + (pericentral program ↓)	**PC-anchor** pericentral only	**Dual** both — co-expression
PC –	**Null** neither	**PP-anchor** periportal only

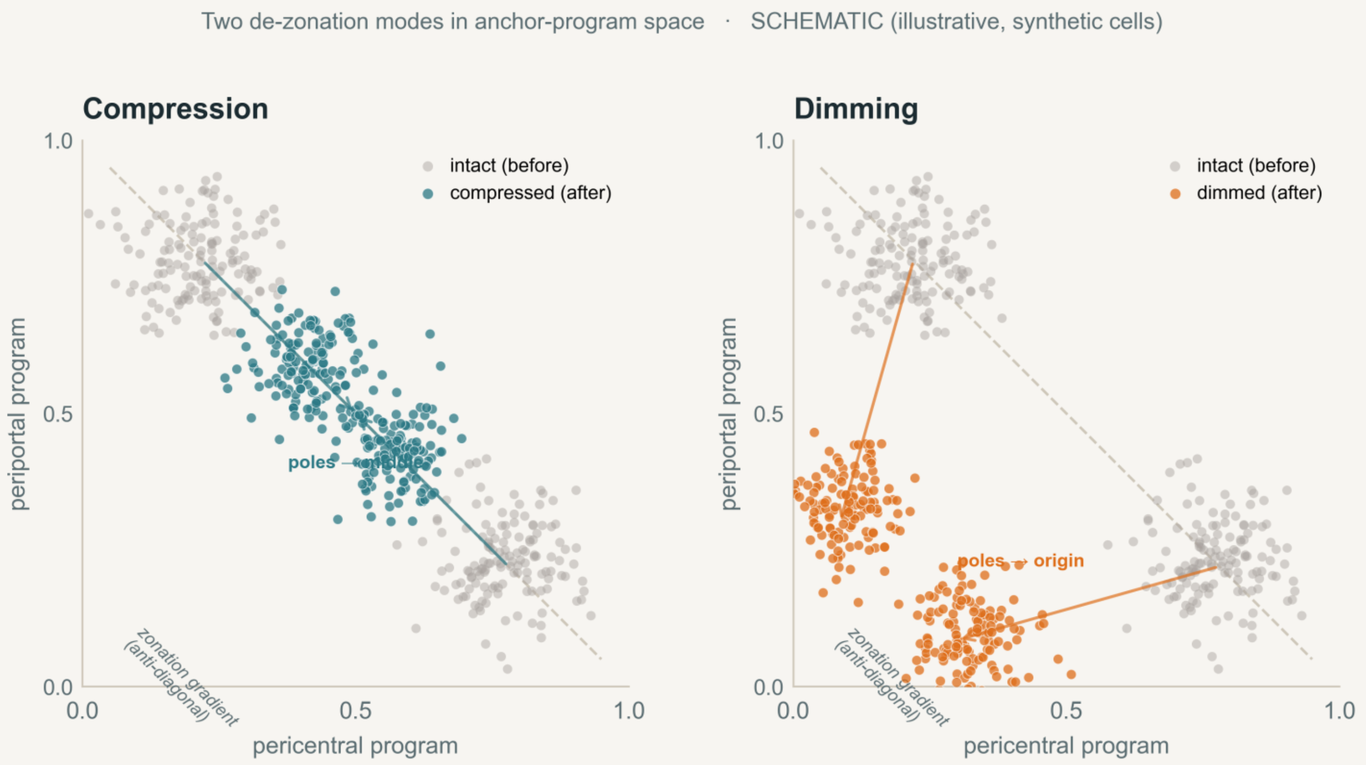
THE FOUR MOVES

Depletion an anchor box ↓	**Co-expression** the dual box ↑
Turn-off the null box ↑	**Composition shift** PP : PC ratio moves

The moves are dissociable — an emptying anchor box (depletion) and a filling null box (turn-off) are opposite directions; neither implies the other.
A single z-score or correlation collapses all four into one number — only counting molecules tells them apart.

Beyond the boxes — zonation is a gradient

Zonation is a continuous gradient — cells strung on an anti-diagonal. It can weaken two ways **without any cell changing class:**



Compression — *shape*

Balance drifts to the middle; the poles' contrast shrinks.

Dimming — *level*

The pericentral detox program declines in level — same balance, lower volume.

Raw-count anchor classification

raw UMI counts → down-thin to 1,500 → **marker ON if ≥ 2 UMI** → classify nucleus (e.g. PC) → fraction per donor

Raw counts, not SCT

A single molecule may be ambient — absolute detection is what matters.

Depth-matched[1]

Down-thin every nucleus to 1,500 UMIs so depth can't drive detection (stable at 1k / 1.5k / 3k).

≥2 UMI, not ≥1

Kills ambient soup — apparent co-expression 7–10% → 0.2–0.5%.

Donor is the unit (~47)

No pseudoreplication. Robust on 6 marker sets (2–1,637 genes, incl. Paper 2's own).

[1] Depth-matched down-sampling: Amezquita et al., Nat. Methods 2020. · Raw-extraction integrity: 9/9 sanity checks pass (integer UMIs; 69,426 hepatocytes match the paper).

Results

what the counts actually say

No depletion signal across matched biopsies

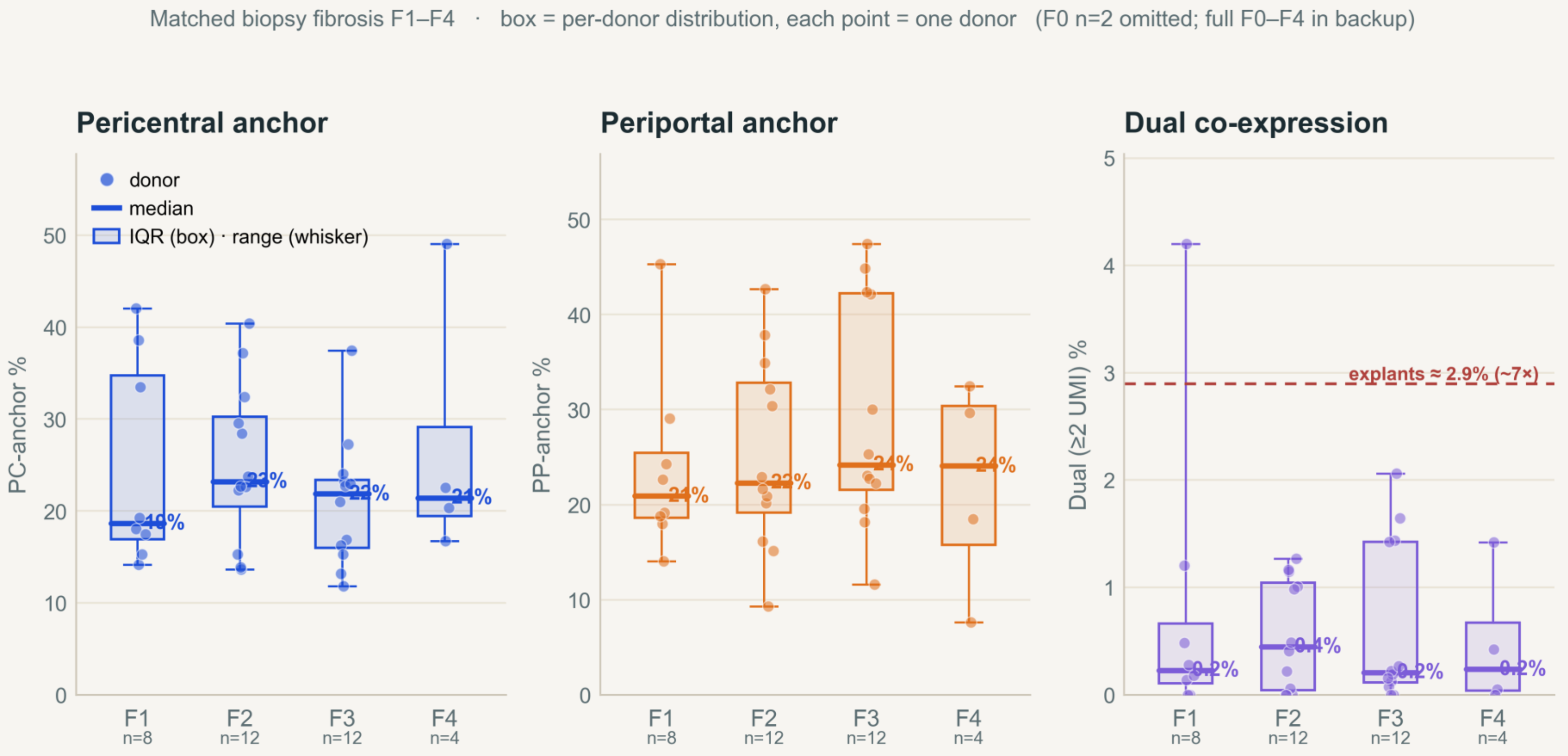


Figure 2. Pericentral / periportal anchor and dual co-expression across matched biopsy F1–F4 (box = per-donor spread; F0 n=2 omitted — in backup).

No depletion.

PC & PP anchors flat across F1–F4.

No co-expression.

Dual stays ~0.4% (vs 2.9% in explants).

Null & PP:PC flat too.

44/36/39/39%; ratio ≈ 1.1.

→ no transcriptional de-zonation signal.

Spread, the equivalence bound (> ±19 pp excluded) and the 10-confounder audit — all in backup.

Cells stay spread across the zones — no collapse

Each panel: where hepatocytes fall on the pericentral ↔ periportal axis.
A spread of positions = zonation present; cells bunched into one pile = collapse.

Per-cell zonal balance by stage — the distribution barely moves across biopsy F1–F4 (dashed = F1 reference); only the confounded explant collapses to one pole

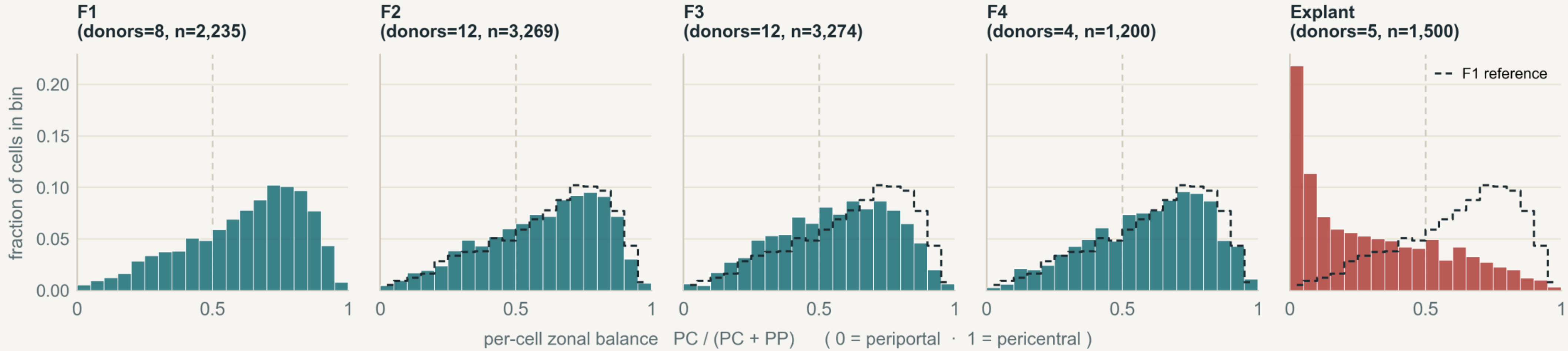


Figure 3. Per-cell zonal balance by stage (down-thinned to 1,500 UMIs, ≤300 nuclei/donor).
Dashed outline = the F1 distribution, repeated on each panel: biopsy F1–F4 barely move; only the explant collapses to the periportal pole.

Across biopsy F1–F4 the mass stays spread — the middle [0.4–0.6) drifts only 22 → 24 → 28 → 26% (peak F3, then reverts), non-monotone and mild. **Only the confounded explant collapses to a single (periportal) pole.**

Per gene, nothing usable moves

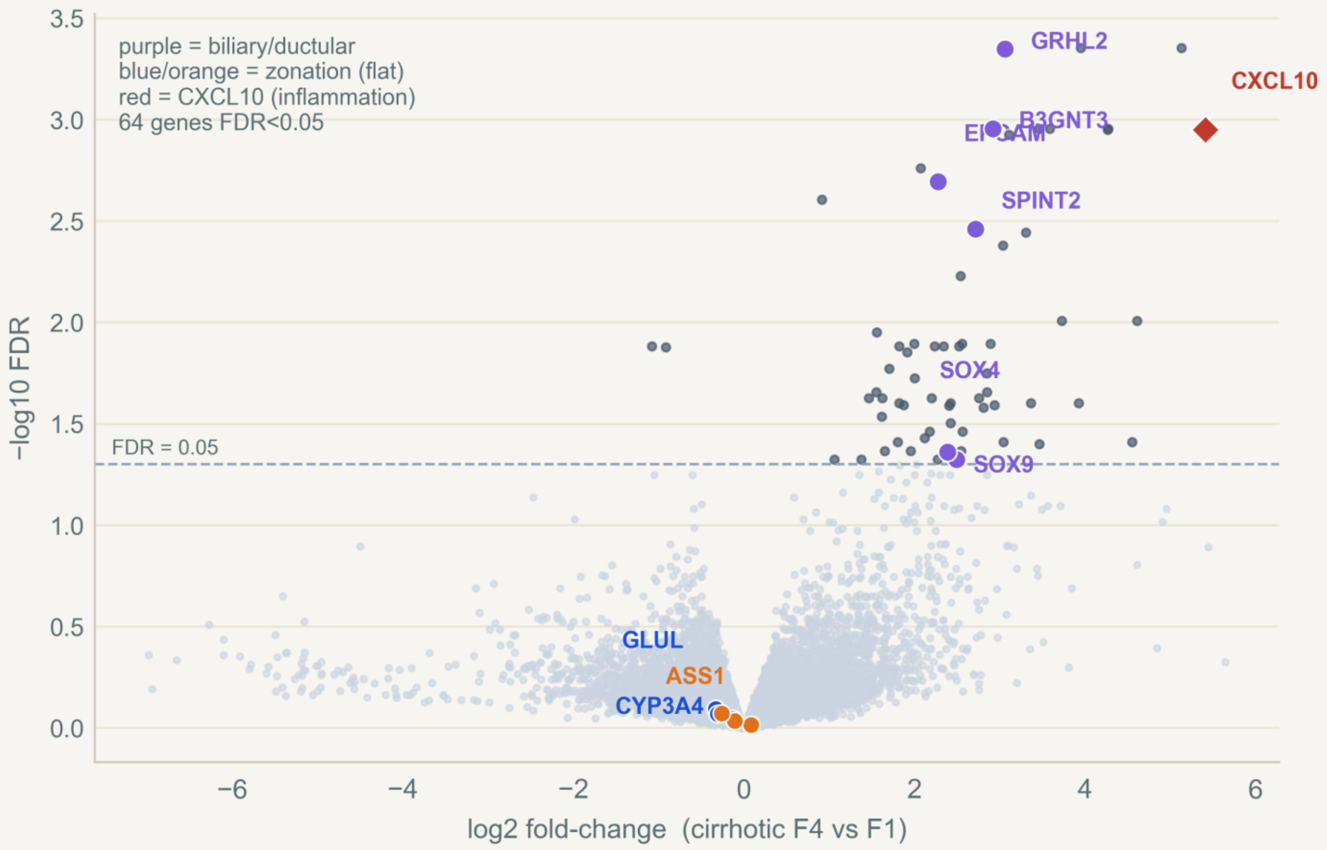


Figure 4. Genome-wide pseudobulk differential expression, cirrhotic F4 vs F1 (edgeR).

64 of ~21,000 genes pass FDR[1]

— mostly biliary / ductular.

Zonation genes flat (GLUL FDR 0.80).

No large single-gene program.

But a weak program, spread thinly across many genes, hides from per-gene FDR

[1] *Pseudobulk: Squair et al. 2021; edgeR / TMM: Robinson, McCarthy & Smyth 2010; Robinson & Oshlack 2010.*

The biliary signal: a compositional confounder? — a lead

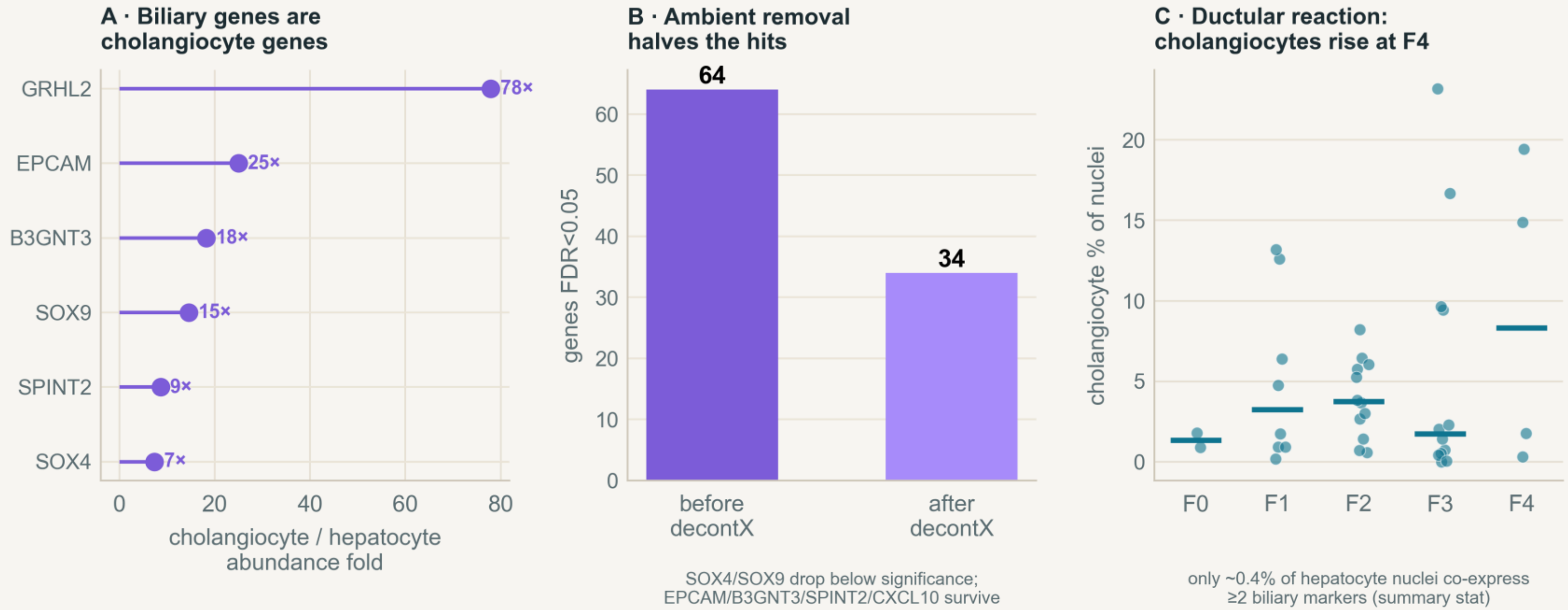


Figure 6. Source attribution — cross-lineage abundance, decontX ambient-RNA removal, and the F4 ductular reaction.

STRONG	Biliary genes are cholangiocyte genes; hepatocyte co-expression ~0.4%
SUGGESTIVE	decontX[1] halves the hits — an ambient contribution, not proof
OPEN	EPCAM / SPINT2 / B3GNT3 survive — a rare intrinsic state not excluded; CXCL10 a separate inflammatory lead

[1] *Ambient-RNA removal: decontX — Yang et al., Genome Biol. 2020. Source attribution is a lead, not a closed verdict.*

Pericentral detox dims — identity stays

Cells keep their class — but within pericentral nuclei the detox program's level falls with fibrosis

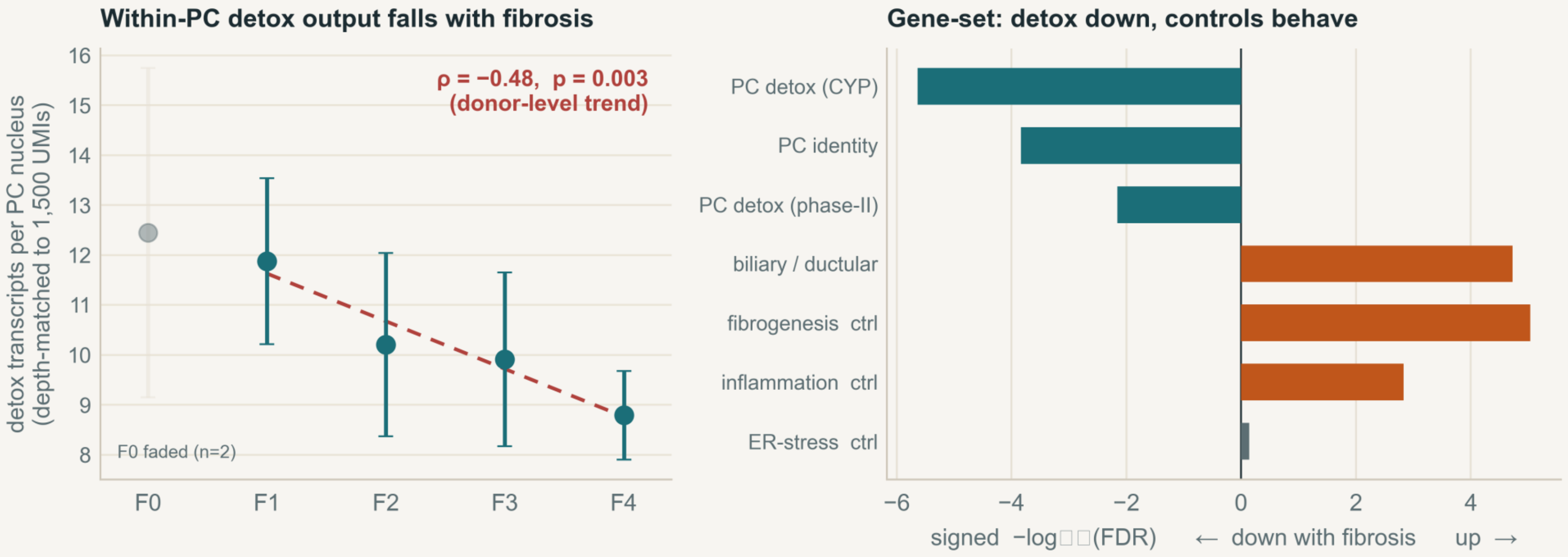


Figure 5. Left — within-PC detox output[1] (depth-matched transcripts per PC nucleus) across fibrosis. Right — gene-set programs[2]: PC detox / identity down, biliary + fibrogenesis up, validating controls behaved.

Like a radio: *the genre held (cells keep identity); the volume dropped (the program dims).* A relative decline (transcripts within a fixed budget), not proven absolute molecule loss; a donor-level trend; could partly reflect a shift among PC subzones. No single gene clears per-gene FDR — only the coordinated set + the per-cell trend.

[1] Within-PC detox output: depth-matched to 1,500 UMIs, donor-level Spearman (ρ=−0.48, p=0.003); measured on detox genes (CYP2E1, CYP1A2, ADH4…) disjoint from the GLUL/CYP3A4 identity anchors — so the anchor fraction stays flat while the module dims. [2] Gene-set: camera — Wu & Smyth 2012; ROAST — Wu et al. 2010; GSEA — gseapy.

CONCLUSION

Zonal identity holds — the pericentral detox program dims

01 **Cells keep their zonal genomic identity — no de-zonation signal**

02 **The one real change: the pericentral detox program dims**

03 **The dramatic genomic trajectory possibly tracks acquisition, not disease**

We correct one evidence leg *— the snRNA marker-correlation; the paper's imaging and protein staining are untouched. The biliary signal is a separate compositional lead.*

Scope — a "PC-like program," not lobular location; a weak coordinated gene-set program is not excluded; F4 is only 4 donors.

Are you overturning the paper? No

- We re-tested only the snRNA-seq transcriptional zonation/plasticity evidence.
- We did NOT test imaging, protein staining, organoids, or 3D architecture.
- Claim: matched biopsy transcriptomes do not show progressive zonation loss.
- We agree the strong signal is an end-stage phenomenon.

Null and the PP : PC ratio are flat too

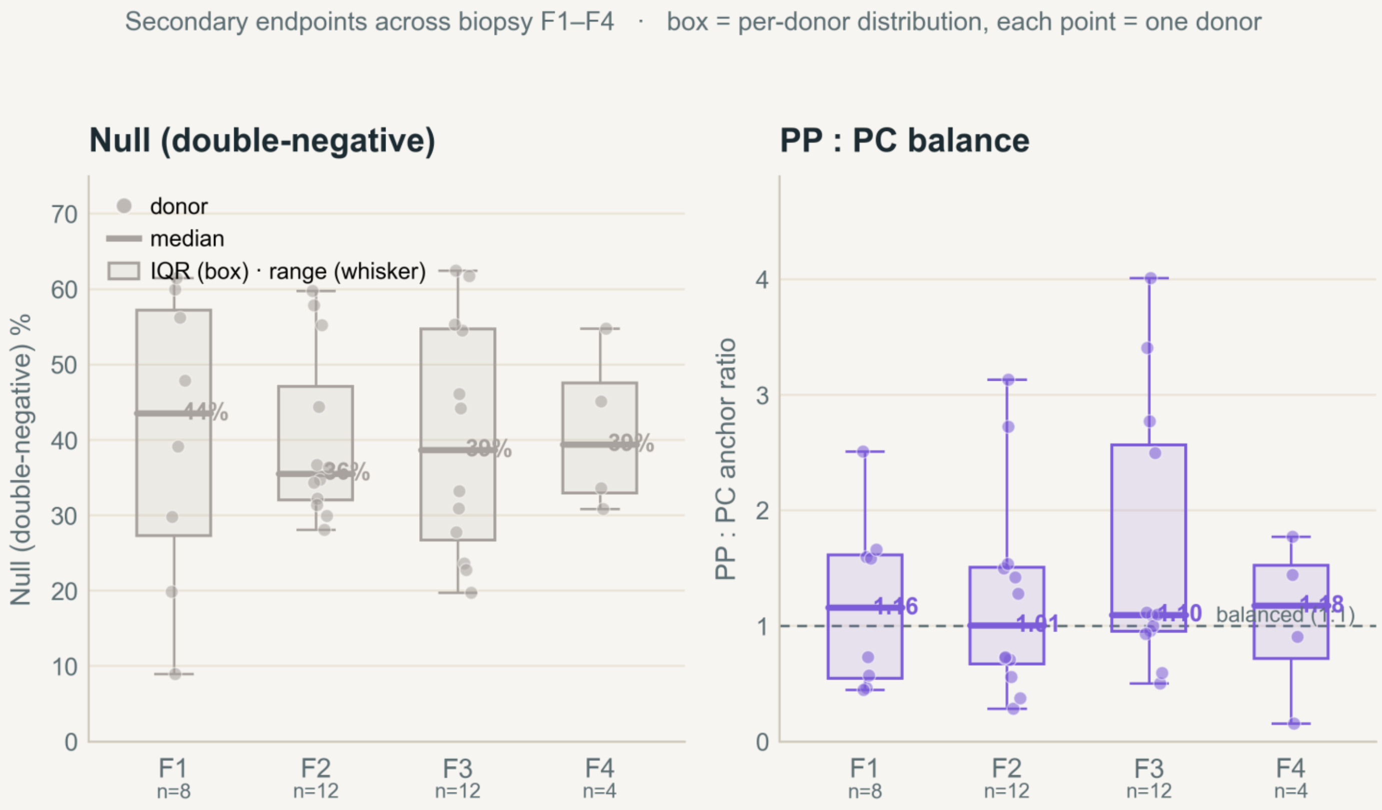


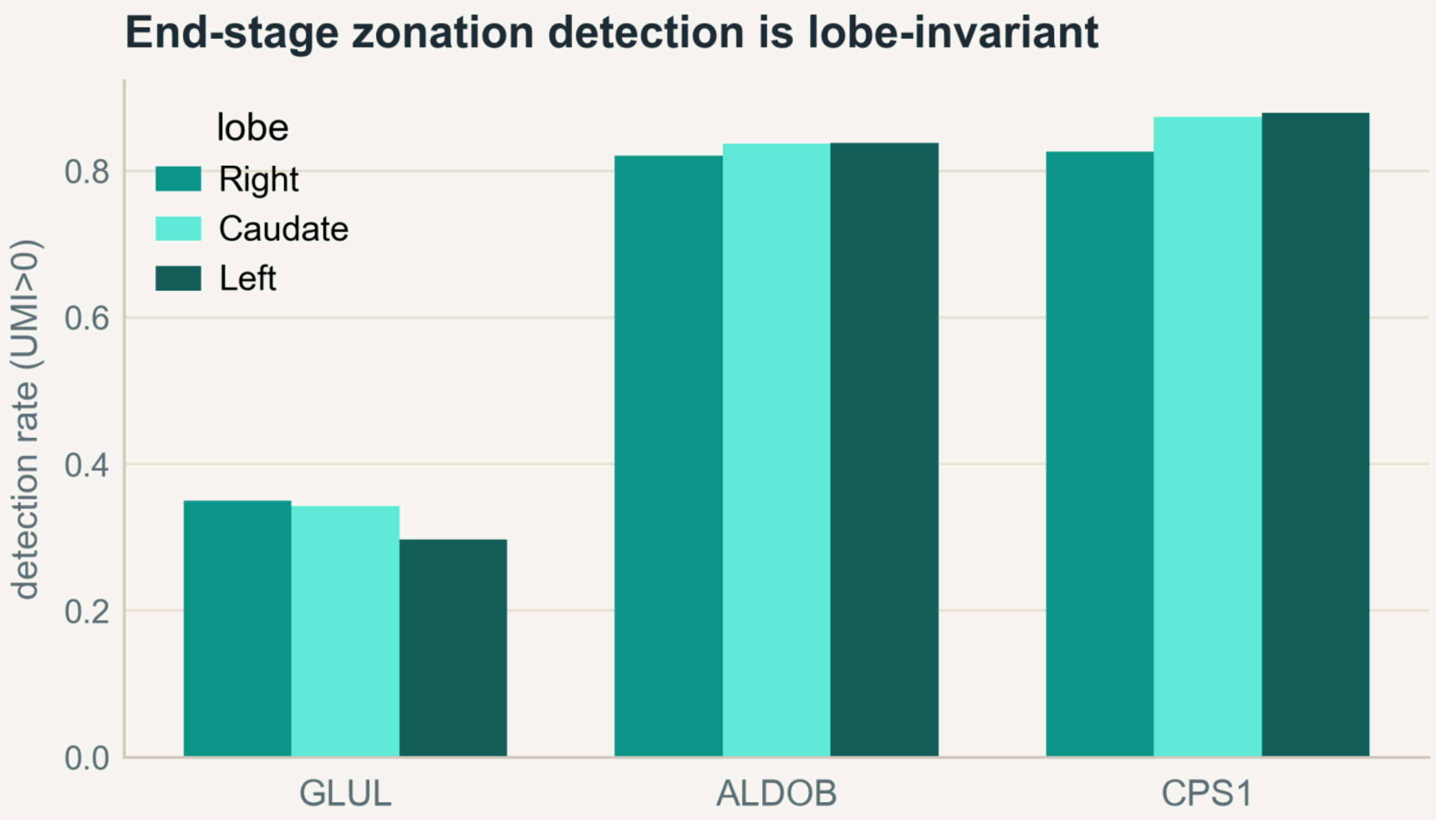
Figure 7. Null (double-negative) fraction and the periportal : pericentral anchor ratio across biopsy F1–F4 (box = per-donor spread; each point = one donor).

Null (double-negative): ≈ 44 / 36 / 39 / 39% — no rise across stages.

PP : PC ratio: ≈ 1.16 / 1.01 / 1.10 / 1.18 — hovers around the balanced 1 : 1 line.

Same large person-to-person spread as the anchors, same absence of a stage trend.

Right, caudate and left lobes all agree

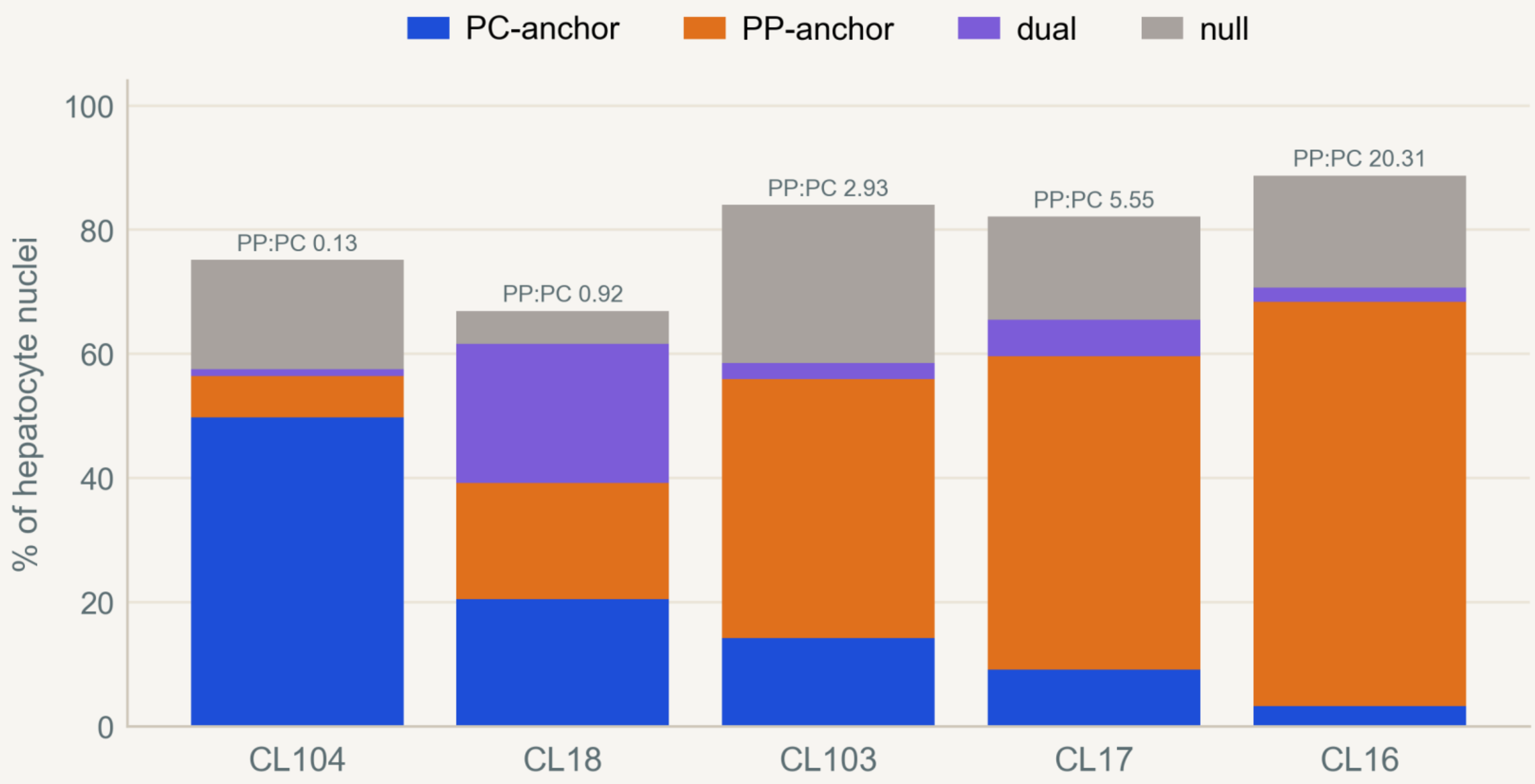


Detection matches across all three lobes.

Lobe does not explain the signal; explant-vs-biopsy acquisition still does.

Figure 8. Zonation-marker detection across Right / Caudate / Left lobes in end-stage explants.

Five organs, five phenotypes



PC-anchor 3% → 50%; PP:PC 0.13 → 20.

Dramatic but NOT one program — pooling manufactures a uniform "collapse."

Figure 9. Per-explant anchor fractions — end-stage is heterogeneous, not one coherent program.

Sensitivity grid + equivalence bound

Stable across every variant:

anchor genes, periportal rule, ALDOB in/out, CPS1-based, 1 vs 2 UMI.

No depletion · no dual rise · no turn-off — in all.

Bound (TOST): large shift (~19 pp) excluded; subtle drift ≤10 pp not (F4 n=4).

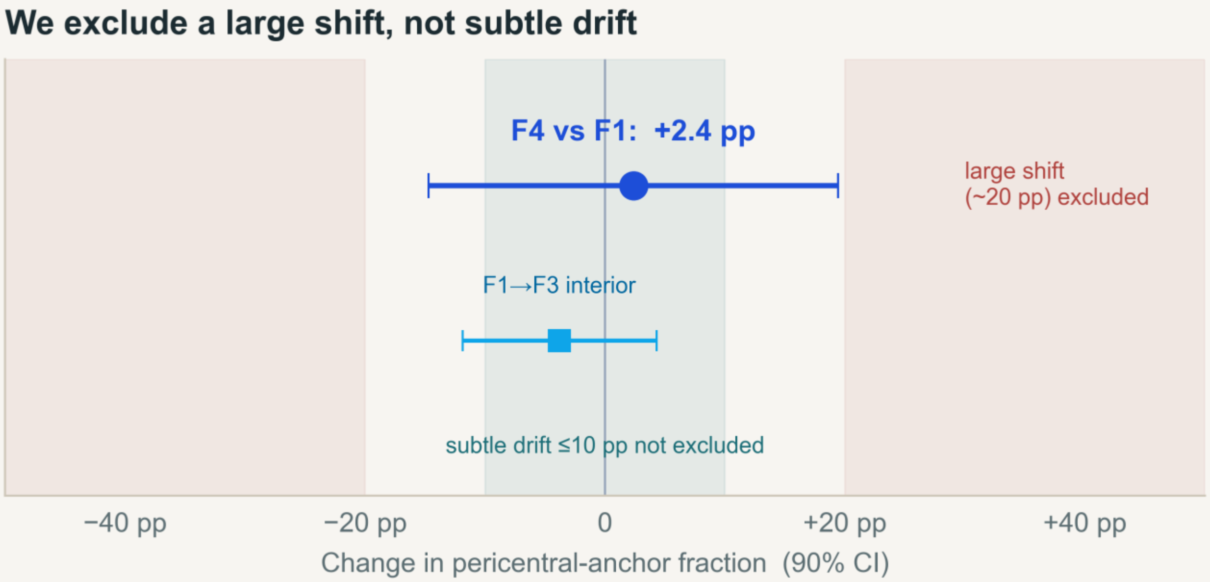


Figure 10. Equivalence bound on the F4-vs-F1 pericentral-anchor change.

Confounder battery — the numbers

Confounder	What we did (test · numbers)	Verdict
Tissue source	organ-cube / explant ends are not acquisition-matched to needle biopsy	excluded → biopsy-only F1–F4
Procurement stress	IEG+HSP across 6 lineages; hep 18× ≈ endothelial 18× (no zonation)	organ-wide handling, not zonation
Sequencing batch	run × source Cramér's V = 0.84; run × F (biopsy) = 0.40	ends confounded; biopsy estimable
Lobe (R / C / L)	marker detection across right / caudate / left lobes in explants	lobe-invariant
Sequencing depth	binomial down-thin to B = 1,000 / 1,500 / 3,000 UMIs	PC-share flat at every B (SD 0.006–0.010)
Ambient RNA ("soup")	corr(ALB share , dual fraction) across 38 donors = +0.04	does not drive co-expression
Cholangiocyte mis-annotation	$KRT19^{+}$ ≤ 0.001, $EPCAM^{+}$ ≤ 0.003 in every anchor class	not contaminating cholangiocytes
Ploidy / complexity	nFeatures by stage: 2218 / 3063 / 3169 / 2641 / 1984	non-monotone — does not track stage
Depth-match discard	3.5–7.8 % dropped below B, uniformly lower-detection	depth-driven, not PC-biased
Clinical covariates	partial corr(detox , F | Age) = −0.32; Age–F = +0.43	demographics do not explain trends

Why raw counts, why pseudobulk

- Raw counts: marker co-detection + ambient sensitivity need molecules, not SCT residuals.
- Pseudobulk: cells from one donor are not independent — sum per donor, then edgeR (Squair 2021).
- edgeR: filterByExpr → 21,022 genes; TMM; NB quasi-likelihood GLM, robust dispersion; common BCV 0.405.
- Primary signal persists in DIRECTION with a run covariate and within a single run; significance attenuates with n (not a batch artifact).

Does decontX prove contamination? No

- decontX models a native + ambient RNA mixture and subtracts the estimate.
- Hits 64 → 34; SOX4/SOX9 drop below significance → supports an ambient contribution.
- But it can remove true hybrid signal OR leave residual contamination.
- Source attribution is a lead, not a closed verdict; CXCL10 is a separate candidate inflammatory signal.

Ten pre-specified sets — the detox program goes down

Gene set	Zone / program (identity)	n	Dir	camera FDR
xenobiotic_CYP	PC detox — phase-I (CYP)	13	↓	2×10^{-6}
detox_phase2	PC detox — phase-II (UGT/GST/SULT)	16	↓	7×10^{-3}
pericentral_anchors	PC identity (Paper 2 landmarks)	20	↓	1.5×10^{-4}
periportal_anchors	PP identity (Paper 2 landmarks)	20	↑	1.5×10^{-2}
urea_cycle	PP program (nitrogen)	9	↓	0.12 (ns)
bile_acid_lipid	mixed metabolism	20	–	0.72 (ns)
cholangiocyte_ductular	biliary / bile-duct markers	10	↑	1.9×10^{-5}
CTRL_interferon	inflammation (+control)	20	↑	1.5×10^{-3}
CTRL_EMT	fibrogenesis / scar (+control)	20	↑	9×10^{-6}
CTRL_ER_stress	unfolded-protein resp. (−control)	20	–	0.72 (ns)

Donor-level, biopsy F0–F4, limma::camera (GSEA agrees). Pre-specified, no fishing: PC/PP identity = Paper 2's own landmark genes; CYP / detox / urea / bile = KEGG·HGNC cores; biliary = plasticity.txt + markers; controls = MSigDB Hallmark cores. The validating controls behaved (fibrogenesis & inflammation ↑, ER-stress flat) — proof the test detects real biology, not noise. The pericentral detox + identity programs fall; the see-saw (PP ↑) and biliary ↑ are why we adversarially controlled for composition (it survived).

Gene-set testing, in one breath each

Pseudobulk — sum each donor's hepatocyte counts into one profile per donor (donor = unit of inference).

CPM — each gene as a share of the donor's total counts; simple, but compositional (a few surging genes shrink everyone else's share).

TMM — the scaling factor is taken from the bulk of genes after trimming the extreme movers, so surging genes can't set it → composition-robust.

Per-gene DGE — test each gene on its own for a fibrosis trend (edgeR); misses a weak shift shared by many genes.

Gene-set test — test a whole pre-specified set at once. camera = competitive (more shifted than the rest of the genome?; Wu & Smyth 2012); ROAST = self-contained (does the set move against zero?; Wu et al. 2010).

Within-PC detox output — detox transcripts per pericentral nucleus after equalizing read depth: per-cell, relative to the cell's own output, immune to between-cell composition.

Every de-zonation route, mapped to its signature

Mechanism	Signature	Biopsy F0→F4 (donor-median)	Verdict
Pericentral depletion	PC-anchor fraction ↓	36 / 19 / 23 / 22 / 21 %	flat → no depletion
Periportal depletion	PP-anchor fraction ↓	20 / 21 / 22 / 24 / 24 %	flat → no depletion
Dimming	within-PC level ↓ (magnitude)	11.9 → 8.8 detox UMI/nuc (F1→F4); ρ = −0.48	REAL — gene-set + within-PC confirm (CYP FDR 2e-6)
Co-expression	dual (≥2 UMI) fraction ↑	~0.0 / 0.2 / 0.4 / 0.2 / 0.2 %	ambient soup → no
Gradient compression	per-cell balance → middle	mild, non-monotone (peak F3)	gradient present
Turn-off	null (double-neg) fraction ↑	34 / 44 / 36 / 39 / 39 %	flat → no turn-off
Composition shift	PP : PC anchor ratio	0.62 / 1.16 / 1.01 / 1.10 / 1.18	~flat → no shift
Induction	program level ↑	flat in biopsy; rises only end-stage	explant-only

Each row is a distinct way zonation could fail, mapped to its signature on depth-normalized counts. The structure-redistribution routes (cells crossing boxes) are all flat — cells keep their identity. The one magnitude route, dimming, is REAL: the pericentral detox program quiets, confirmed by gene-set + within-PC testing.

Conclusions · limits · next steps

Conclusions

- Full healthy→end-stage trajectory is source-confounded.
- In biopsy F1–F4 cells keep zonal identity — no de-zonation (large shift excluded).
- Two real biopsy-internal signals: a compositional biliary signal (a lead), and a confirmed pericentral-detox dimming (gene-set + per-cell).

Limits

- “PC-like program,” not lobular location — counts can’t see spatial position.
- Imaging / protein / organoid evidence not re-analyzed.
- F4 has only 4 biopsy donors.
- Detox dimming is a RELATIVE decline (not proven absolute) and could partly reflect a shift among PC subzones; self-contained test clears only the CYP core.

Next

- Spatial / independent-biopsy validation of the detox-dimming gradient.
- Leave-one-F4-donor-out DGE.
- Quantitative contamination model for the biliary lead.
- Functional readout of pericentral detox capacity.